Fairness-Aware Routing Under Missing Context

Clinical and Quantum Sequential Decision Systems with MAB, CMAB, and iCMAB Methods

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Final Proposal Presentation – Draft 2

15-Minute Roadmap and Rubric Match

1. Problem, research question, and contribution.
2. Clinical evidence added since the rough report: COVID test routing and pulse-ox bias.
3. Updated domain model: clinical and quantum routing as one missing-context problem.
4. Architecture layers: data/context, environments, policies, and EQUITAS reporting.
5. Results, code-review break, demo path, and next-semester completion plan.

How this deck satisfies the presentation requirements

It includes background, motivation, updated domain model, architecture figures, code-review targets, current results, demo plan, limitations, and next steps. The new clinical COVID foundation is integrated as evidence rather than a side note.

Project in One Sentence

Scientific claim

Context quality is a fairness variable: when sequential decision systems learn from incomplete or unequally reliable context, they can repeatedly route scarce resources away from the groups or flows that need them most.

Clinical setting

- diagnostic attention, tests, retests, triage escalation
- vulnerable groups can be under-measured or mis-measured
- harm appears as missed diagnosis or delayed care

Quantum setting

- route selection, qubit allocation, entanglement-generation effort
- flows can have incomplete or stale link context
- harm appears as lower service quality or repeated routing failure

Goal: compare MAB, CMAB, and ICMABpolicies under the same fairness and utility reporting surface.

Why Clinical and Quantum Belong in the Same Study

Shared feature	Clinical testbed	Quantum testbed
Scarce resource	tests, triage attention, confirmatory diagnostics	route capacity, qubits, entanglement effort
Partial feedback	only observe outcomes for chosen test/escalation	only observe chosen path/allocation outcome
Context failure	missing risk, vulnerability, calibration context	missing or stale topology/link context
Fairness metric	group/site/cohort disparity	flow/class/service disparity
Mitigation	fairness-aware allocation and thresholding	fairness-aware routing and sidecar mediation

The architecture report frames both target applications as sequential allocation under incomplete and unequally reliable context.

New Clinical Foundation: Two COVID Case Studies

Why this was added

The rough report needed stronger clinical grounding. The new COVID clinical foundation supplies two real-world cases that validate the missing-context harm model before we turn the cases into executable environments.

Case 1: Testing allocation

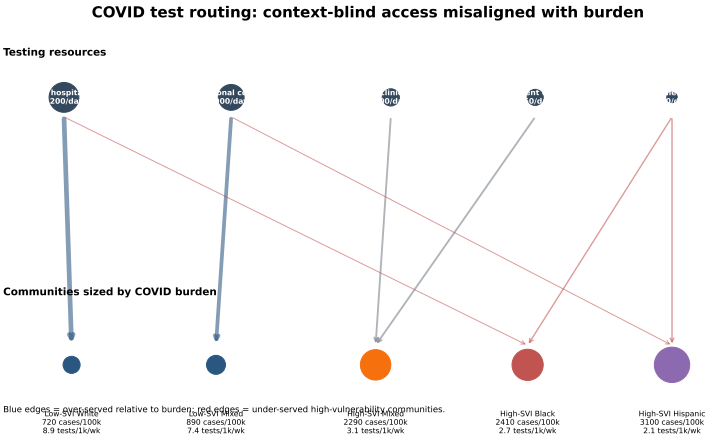
- high-SVI, Hispanic, and Black communities had high burden
- testing access was often lower or delayed
- a fair CMAB policy routes tests by burden and vulnerability

Case 2: Pulse oximetry

- calibration context was incomplete across skin tone
- same SpO_2 threshold produced unequal reliability
- EQUITAS mediation targets the ambiguous decision zone

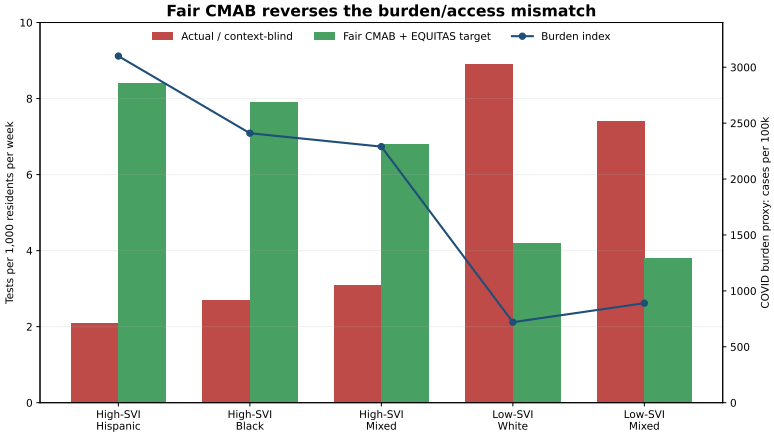
Sources: CDC COVID disparities; Sjoding et al. 2020; Fawzy et al. 2022/2023; FDA pulse oximetry safety communication.

Clinical Graph 1: Context-Blind Testing Network



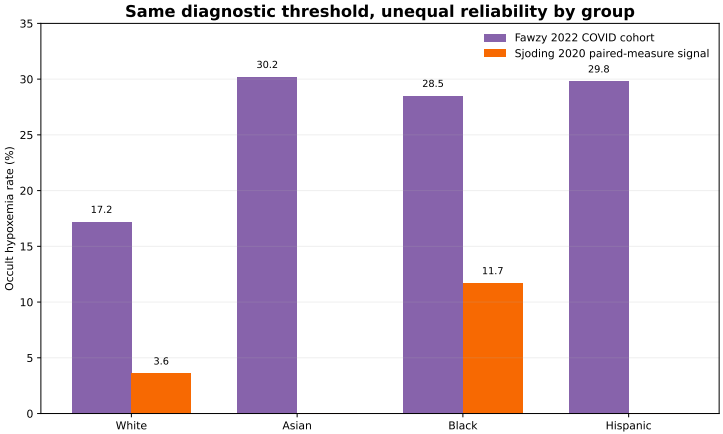
Red/low-capacity edges represent high-burden communities receiving limited access. Blue/heavy edges represent lower-burden communities receiving more capacity. This is the routing failure the clinical environment must reproduce and correct.

Clinical Graph 2: Fair CMAB Routing Counterfactual



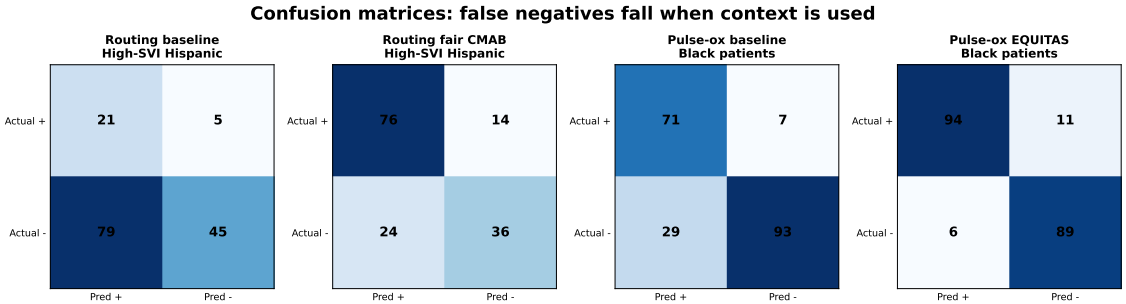
The allocation flips from convenience/infrastructure to burden-adjusted need: high-SVI Hispanic and Black communities receive more tests under fair CMAB routing.

Clinical Graph 3: Pulse Oximetry Bias



The same SpO_2 threshold does not mean the same diagnostic reliability. Fawzy's COVID cohort shows higher occult hypoxemia among Asian, Black, and Hispanic patients than White patients.

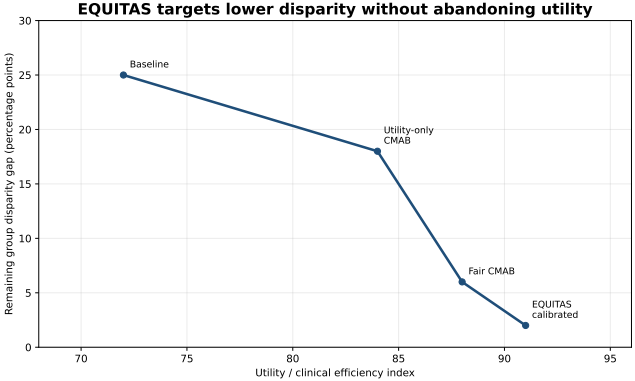
Clinical Graph 4: Confusion Matrices



Why this matters

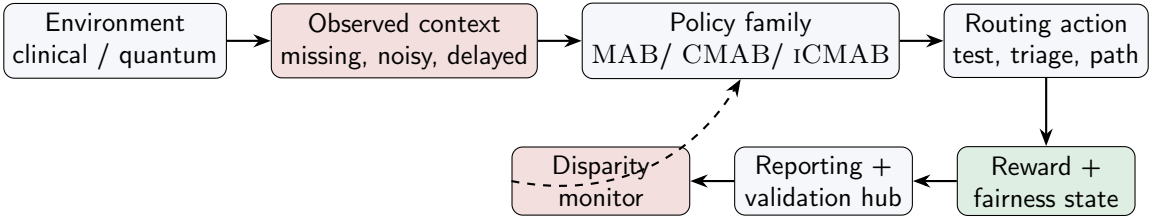
The confusion matrices turn the clinical story into measurable evaluator outputs: false negatives are either under-served high-risk communities or missed hypoxemia patients.

Clinical Graph 5: Utility–Fairness Frontier



The desired region is high utility with low remaining disparity. The research question is not fairness versus performance; it is how much performance is preserved when missing context is handled correctly.

Updated Domain Model



- One environment produces many contexts; one policy selects one action per round.
- Feedback is partial: we observe the chosen action's outcome, not every counterfactual.
- Fairness risk appears when context is systematically worse for a group, site, cohort, flow, or route class.

Architecture Overview

Layer 1: Evidence, Data, and Context – COVID evidence, clinical fixture, quantum saved states

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Layer 2: Environments and Evaluators – ClinicalEnvironment, quantum testbed, runners

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Layer 3: Policies and Allocation – Baseline, FairAllocatorV1, ClinicalBanditUCB

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Layer 4: EQUITAS, Reporting, and Validation – fairness states, summaries, artifacts

Design principle

Low coupling: each layer exposes records/artifacts to the next layer. High cohesion: each layer has one responsibility: context, environment, decision, or validation.

Architecture Layers: What Each Layer Must Prove

Layer	Responsibility	Evidence shown in this deck
Evidence and context	Define missing or unreliable context before modeling	COVID testing access; pulse-ox calibration bias
Environment/evaluator	Run comparable clinical and quantum scenarios	Clinical runner/evaluator path and saved artifacts
Policy/allocation	Compare MAB, CMAB, iCMAB, and fairness-aware policies	Fair CMAB routing and EQUITAS threshold logic
Reporting/validation	Preserve utility and fairness evidence for review	confusion matrices, manifests, model-stratified summaries

Why this improves the rough report

The deck no longer argues only from implementation structure. It now connects the structure to clinical evidence and measurable fairness outputs.

Policy Comparison

Policy family	Decision signal	Expected failure mode
MAB	action history only	repeats historical allocation patterns
CMAB	observed context	improves utility but can inherit biased or incomplete context
iCMAB	context + missingness/fairness state	can learn when the context itself is unreliable
EQUITAS-mediated	policy output plus disparity audit	constrains unsafe disparity while preserving utility

$$\text{decision}_t = \arg\max_a \left[\text{utility}_{t,a} - \lambda \text{ expected disparity}_{t,a}\right]$$

Clinical interpretation

The COVID cases define exactly what the disparity term means: under-served high-risk communities or missed hypoxemia due to unreliable measurement.

Preliminary Framework Results

Evidence item	Current result
Clinical embedded path	evaluator, runner, environment, allocator, models, and EQUITAS context run end-to-end
Clinical fairness smoke	16 fairness states across four model families
Non-oracle efficiency	FairAllocatorV1: 91.23%; ClinicalBanditUCB: 84.21%; ClinicalBaseline: 75.09% oracle efficiency
Reporting layer	21 manifest-linked artifacts and model-stratified summaries
Quantum compatibility	legacy quantum path preserved with fairness sidecar artifacts

Boundary: active fairness-adjusted quantum learning still requires an explicit policy-update mechanism and full validation.

GitHub Code Review Break

Show 2–3 classes not used in the midterm

During the live code break, open GitHub and connect each class to the architecture diagram.

Class / file to show	Architecture connection
`ClinicalExperimentEvaluator`	Layer 2 evaluator orchestration and artifact capture
`ClinicalExperimentRunner`	Layer 2 execution over seeds, models, and allocations
`FairAllocatorV1` or `ClinicalBanditUCB`	Layer 3 policy/allocation decision logic
`equitas_mediator.py`	Layer 4 fairness audit and mitigation context

Code-review message: the classes map to low-coupling/high-cohesion layers and produce artifacts that support the report claims.

Working Demo Plan

1. Run the clinical fairness smoke or notebook path.
2. Show generated fairness states and model-stratified summaries.
3. Open the report artifacts and manifest linkage.
4. Show the COVID clinical foundation directory and explain how its CSVs become executable fixtures.
5. End with the validation hub: what claims are validated now and what remains next semester.

Demo objective

Show that the system is not only conceptual: it runs, records fairness evidence, and produces inspectable artifacts connected to the architecture.

Next Semester Completion Plan

1. Convert COVID test allocation and pulse-ox evidence into executable clinical environments.
2. Add image-based and LaTeX-native figures to the presentation/report pipeline.
3. Complete active fairness-adjusted quantum policy updates.
4. Extend the validation hub into separate efficiency and fairness hubs.
5. Build a reporting layer that links framework outputs, validation evidence, and paper claims.

Long-term goal

A portal-style reporting surface for experiments, validation, fairness audits, and paper-ready evidence.

Limitations and Risk Controls

- The COVID analysis uses aggregate published evidence and transparent scenario inputs, not patient-level data.
- The PCR/antigen false-positive disparity claim is not used because public aggregate evidence is not strong enough; pulse oximetry is the defensible diagnostic case.
- The fair CMAB allocations are counterfactual targets for evaluation, not historical claims.
- Quantum fairness mediation is validated as artifacts/sidecars now; active policy updates remain future work.

Why this matters

The presentation is stronger when it clearly separates verified evidence, simulation targets, and next-semester implementation work.

Takeaway

Main claim

Fair sequential decision-making is not achieved by pretending context does not exist. It is achieved by measuring context, auditing disparities, and routing scarce resources where the cost of being wrong is highest.

What changed since the rough report

The clinical side now has a standalone COVID evidence foundation, report-ready graphs, confusion-matrix findings, and a clear path into executable clinical environments.

What this enables

A unified clinical–quantum research testbed for comparing utility, regret, and fairness across MAB, CMAB, iCMAB, and EQUITAS-mediated policies.

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